```
R version 4.5.2 (2025-10-31) -- "[Not] Part in a Rumble"
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Platform: aarch64-apple-darwin20

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  Natural language support but running in an English locale

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.82 (8556) aarch64-apple-darwin20]

[History restored from /Users/syjean/.Rapp.history]

> devtools::install_github("ccolonescu/POE5Rdata")
Downloading GitHub repo ccolonescu/POE5Rdata@HEAD
── R CMD build ─────────────────────────────────────────────────────────────────────────────────────────────────────────────────
─  checking for empty or unneeded directories/xb2yzt612993njbck79mmj8m0000gp/T/Rtmpmm9rSG/remotes1bc116f5f90a/ccolonescu-POE5Rdata-f23f7b5/
DESCRIPTION' ...
─  building 'POE5Rdata_0.1.0.tar.gz' (538ms)
   building 'POE5Rdata_0.1.0.tar.gz' (538ms)
> library(POE5Rdata) #activates the package "POE5Rdata"

Attaching package: 'POE5Rdata'

The following object is masked from 'package:datasets':

    euro

> data(andy) #reads the dataset `andy` into the global environment
> ?andy      #shows information about the dataset
starting httpd help server ... done
> head(andy) #displays the head of the dataset `andy`
  sales price advert
1  73.2  5.69    1.3
2  71.8  6.49    2.9
3  62.4  5.63    0.8
4  67.4  6.22    0.7
5  89.3  5.02    1.5
6  70.3  6.41    1.3
> lm(sales~price+advert, data=andy) #a regression model

Call:
lm(formula = sales ~ price + advert, data = andy)

Coefficients:
(Intercept)        price       advert
    118.914       -7.908        1.863

> data(filename)
Warning message:
In data(filename) : data set 'filename' not found
> data(filename)
Warning message:
In data(filename) : data set 'filename' not found
> devtools::install_github("ccolonescu/POE5Rdata")
Skipping install of 'POE5Rdata' from a github remote, the SHA1 (f23f7b56) has not changed since last install.
  Use `force = TRUE` to force installation
> library(POE5Data)
Error in library(POE5Data) : there is no package called 'POE5Data'
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> library(POE5Rdata)
> mod1 <- lm(totalscore ~ small, data = star5_small)
> smod1 <- summary(mod1)
```

```
> smod1

Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-191.63  -50.63   -7.12   42.37  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  918.628      2.536 362.260   <2e-16 ***
small          9.989      4.611   2.166   0.0305 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,	Adjusted R-squared:  0.003071 
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047

> mod2 <- lm(readscore ~ small, data = star5_small)
> smod2 <- summary(mod2)
> smod2

Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-76.123 -21.123  -3.356  15.877 192.877 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  434.123      1.044 416.021  < 2e-16 ***
small          5.466      1.897   2.881  0.00403 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,	Adjusted R-squared:  0.006053 
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032

> mod3 <- lm(mathscore ~ small, data = star5_small)
> smod3 <- summary(mod3)
> smod3

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max 
-145.505  -35.028   -6.505   28.495  141.495 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  484.505      1.677 288.884   <2e-16 ***
small          4.522      3.049   1.483    0.138    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,	Adjusted R-squared:  0.0009992 
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383

> 
> mod4 <- lm(totalscore ~ aide, data = star5_small)
> smod4 <- summary(mod4)
> smod4

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-195.14  -52.14   -6.95   40.95  330.86 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  922.145      2.640 349.250   <2e-16 ***
aide          -1.396      4.437  -0.315    0.753    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05,	Adjusted R-squared:  -0.000752 
F-statistic: 0.09904 on 1 and 1198 DF,  p-value: 0.753

> mod5 <- lm(readscore ~ aide, data = star5_small)
> smod5 <- summary(mod5)
> smod5

Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-77.908 -21.536  -2.908  15.464 191.092 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
```

```
(Intercept) 435.9084     1.0882 400.585   <2e-16 ***
aide         -0.3719     1.8285  -0.203    0.839
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05,	Adjusted R-squared:  -0.0008002 
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389

> mod6 <- lm(mathscore ~ aide, data = star5_small)
> smod6 <- summary(mod6)
> smod6

Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max 
-147.236  -32.236   -7.212   27.788  140.788 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  486.236      1.744 278.731   <2e-16 ***
aide          -1.024      2.931  -0.349    0.727    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019,	Adjusted R-squared:  -0.0007327 
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268

>   library(POE5Rdata)
> data("cex5_small")
> ?cex5_small
> hist(cex5_small$foodaway, col='grey')
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   12.04   32.55   49.27   67.50 1179.00 
> 
> summary(cex5_small$foodaway[cex5_small$college==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   14.44   36.11   48.60   68.67  416.11 
> summary(cex5_small$foodaway[cex5_small$advanced==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   21.67   48.15   73.15   90.00 1179.00 
> summary(cex5_small$foodaway[cex5_small$college==0 & cex5_small$advanced==0])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00    9.63   26.02   39.01   52.65  437.78 
> table(cex5_small$college, cex5_small$advanced)
   
      0   1
  0 574 257
  1 369   0
> 
> hist(log(cex5_small$foodaway), col='grey')
> summary(log(cex5_small$foodaway))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   -Inf   2.488   3.483    -Inf   4.212   7.072 
> 
> mod1 <- lm(log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 
    0, ])

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738 
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16

> 
> plot(cex5_small$income[cex5_small$foodaway > 0],
+      log(cex5_small$foodaway[cex5_small$foodaway > 0]),
+      col="grey")
> 
> abline(mod1, col="blue", lwd=2)
> 
> resids <- residuals(mod1)
> plot(cex5_small$income[cex5_small$foodaway > 0],
+    resids,
+    col="grey")
> 
> 
```